RFS BioConsulting - Emily Destito, Jake Pena, Hayden Spector, Ian Vann

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Complete qualitative evaluation of Photys Therapeutics' potential as a lucrative investment, assessing the company's science, potential, market size, potential risk, and feasibility.

Photys Therapeutics is a company pioneering the ability to apply small molecule binding to modulate and induce phosphorylation-dependent cellular processes. An academic spinoff from the Broad Institute in Cambridge, MA, Photys licenses technology researched by founder *Amit Choudhary* during work at Harvard. Existing therapeutics concerning phosphorylation pathways function as 'off' switches, working to inhibit a given response. Choudhary's Research and Photys's work anticipate completely shifting the landscape of phosphorylation-targeted therapeutics, designed bivalent molecules that can recruit and activate any kinase. The company is currently in a post-sealth, post-series-A sequence of funding, but is still pre-IPO. Photys, based out of Waltham, MA, received significant series-A funding from major VC firms and large Pharma companies, but in this stage of funding, there is likely a significant opportunity remaining for investment in series-B funding and beyond.

Science

The primary technology and therapeutic novel to Photys is phosphorylation-inducing chimeric small molecules (PHICS). This therapy utilizes small molecules to drive kinase to the target protein into proximity and induce phosphorylation in proteins that normally could not undergo this process on their own, because of function or degradation of the protein. Therapies utilizing the new PHICS technology allow for the altering of the structure and therefore function of a protein through the modulating of small molecules to induce either native or neo-phosphorylation. PHICS are small chimeric molecules; these synthetic molecules have a specialized bivalent receptor and make proximity-induced phosphorylation possible. These chimeric molecules are engineered to support highly specified binding to the kinase and then form a subsequent bond with the target protein. PHICS are connected with a chemical linker to allow binding to a specialized kinase and target protein. Initially, these PHICS utilized specifically designed recruitment factors and kinase binder sites to locate the receptor, making this proximity-inducing treatment increasingly challenging to apply to other kinases or on a larger scale.[1]

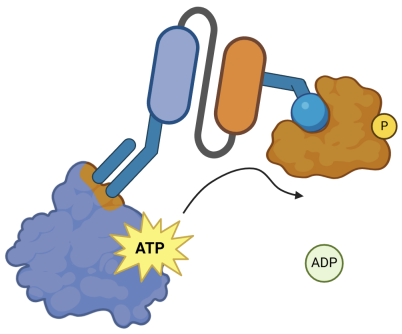


Figure 1: *The model of a PHICS molecule contains a kinase binder, promoting a phosphorylation of the target protein that binds to the other end of the chemical linker.*

However, in 2023, Boundary detailed the use of small molecule inhibitors on the PHICS, which were able to quickly locate the kinases and fleetingly bind, allowing for the ternary complex to form.[2] Using proximity-induced catalysis, kinases can gain a new phosphorylation affinity, inducing phosphorylation at native sites and experiencing neo-phosphorylation. While native phosphorylation refers to phosphorylation that occurs in a protein that is naturally targeted by kinase, neo-phosphorylation refers to artificial phosphorylation that is only possible through utilizing technologies such as PHICS, where a new phosphorylation site is created in the protein through forced proximity of the target protein and kinase (**Fig. 1**).[1]

PHICS binding technology was successfully tested using two different kinases, AMPK and PKC, where chimeric small molecules induced both native and neo-phosphorylation within bromodomain-containing protein 4 (BRD4). This protein is the most studied BET-family protein. BRD4 is highly expressed in different types of tumor cells, where it can act as a protection against immunogenic cell death, making it harder to treat these tumors with traditional chemotherapy or radiation methods. BRD4 is also expressed in normal immune cells as a response to various stresses, making this protein useful in following and predicting disease progression and activity.[3]

PHICS contain recruitment factors and binding sites are highly specific and engineered for a directed phosphorylation-mediated response that does not normally occur. In this case, the chimeric nature of these small molecules refers to the joining of the kinase binder, which targets and activates a specific kinase, and the target protein binder, which directs the kinase molecule in close proximity to the protein of interest. For instance, the kinase binders studied include PF06409577 for kinase AMPK, benzolactam for kinase PKC, and the target binder used for protein BRD4 was (S)-JQ1. These two binders are connected via a flexible chemical linker, consisting of alkyl chains, polyethylene glycol units, and other flexible polymers. When designing PHICS, ternary complex formation needs to be analyzed and compared to BRD4 phosphorylation. This designing process could be accelerated with programs such as AlphaFold and other machine-learning models. These linkers and biological modeling programs are similar to those studied and used during CHEME 2880 Lab sessions, showcasing the relevance of combining different "bio-bricks" in innovative therapeutic technology today while taking advantage of novel programs to streamline this process.[1, 2]

Potential of the PHICS Technology

Currently, therapeutics utilizing kinase act only as inhibitors or "off" switches, which block phosphorylation.[1] Phosphate is one of the most diverse compounds utilized within the cell, such as being used for structure, energy, and information storage.[4] This makes phosphate the "molecular currency" within the cell. Energy within the cell is stored in the phosphoanhydride bonds of ATP and phosphorylated proteins, which play a major role in signaling for various cell functions. Phosphorylation is an example of a post-translational modification of proteins, which is used to regulate a vast majority of cellular functions.[4] Because of this, there is a nearly unlimited number of diseases and other molecular processes that rely on phosphorylation. This gives the PHICS technology extensive potential, with the ability to modify any cellular process or disease through inductive neo -or native phosphorylation in a particular target protein. Potential applications as research and development continues include inducing phosphorylation to fight illness through the release of specific antigens, treating cancer through suppressing tumor proteins or activating apoptosis pathways, treating diabetes through targeting insulin receptors, and fighting obesity through modulating enzymes corresponding to lipid synthesis.

Work in this area has already begun, including various interesting applications and advances of the PHICS technology. In the first iteration of this technology, Choudhary used previously identified kinase activators to bind to the chemical linker. This demonstrated an ability to reconfigure two different serine and threonine kinase enzymes to phosphorylate target proteins, opening up the possibility to expand this technology to different medical applications. Although there is significant potential in the continued advancement and application of this technology, there currently exists a limitation. The method used to find these identified kinases

was extremely limited and required a kinase binder that did not generate an allosteric change, altering enzymatic function.[1]

An interesting application of this technology is combating Alzheimer's disease, where inducing phosphorylation to activate survival pathways in cells can be used to prevent the aggregation of tau proteins, which is a characteristic and driving force of this disease.[5]

A more recent breakthrough includes the ability to rewire tyrosine kinase enzymes utilizing low-affinity chemical inhibitors. These inhibitors bind fleetingly to the kinase, however they allow for a permanent covalent bond to form with the kinase and the target protein. This approach allows for the target of nearly all kinases now, having the potential for a great breakthrough in treatments for hard to combat diseases such as cancer.[6]

Innovation in the Therapeutic Field

Photys developed PHICS to model a synthetic chimera that uses small molecules for targeted protein degradation, similar to proteolysis targeting chimera (PROTAC). These proteolysis-targeting chimeras have a flexible linker that allows for a ligand that recruits the protein of interest (PoI) and another that binds to a degradation enzyme. This process worked as a proof of concept to Photys that proximity can induce naturally limited or nonexistent pathways. This technology has been applied to post-translational modifications, such as Photys' work with phosphorylation. Through this, Photys is not in any direct competition with therapeutics using PROTAC technologies such as Nurix, Arvinas, and Kymera,[7] but rather with other therapeutics that their phosphorylation approach is targeting.

There are no other prominent therapeutics that can induce unspontaneous phosphorylation. This makes Photys' therapeutics a novel solution to countless diseases and disorders that a phosphorylated response can combat. Ultimately, this introduces Photys as a competitor to other existing therapeutics on the market that target many diseases, namely certain cancers, Alzheimer's, and diabetes. Essentially, Photys is introducing PHICS as an entirely new family of therapeutics, placing it against existing small molecule, mRNA, and alternative therapies. For instance, within the diabetes field, PHICS technology will have to compete with insulin injections and alpha-glucosidase inhibitors that have been proven to yield results for an entire patient lifetime. Similarly in cancer and Alzheimer's treatment, Photys will resemble a challenger to the existing pillars of medication.[8]

Applications of Technology and Market

Currently, most therapeutics that use phosphorylation are inhibitors, but there are a few blockers on the market. Of the FDA approved inhibitors, costs range from $2400-$528,000 per year per person, with most of them around $250,000.[9] Blockers range from $144,000-$324,000 per year per person, with a mean monthly cost of $18,000.[9] The average cost of chemo is $27,000 for a six month course, making it a much cheaper alternative to protein kinase blockers and inhibitors.[10] Additionally, about two thirds of the inhibitors available treat orphan drugs, not making them a huge money maker.[9] The technology Photys is working on is more complex than blockers and inhibitors, meaning the drugs developed from their research would probably be upwards of $250,000 per year per person on average. The best target for these therapies would be diseases where there are no current cures or ways to prevent decline.

One such disease is Alzheimer's disease, which is a neurodegenerative disorder that is the most common cause of dementia. Tau proteins phosphorylate to regulate the strength of bonding in microtubules, which form the scaffolding for the neuron network in the brain.[11] When tau

proteins clump together and get tangled up, it leads to neurons being disrupted. This occurs in about 75% of all dementia cases.[12] The most common post-translational modification of this protein has to do with phosphorylation.[13] Mice with genetic modifications meant to mimic human patients with genetic mutations of tau were treated with a phosphorylation inhibitor cocktail, and monitored the amount of tau accumulation. The mice model proved that the tau accumulation was a driving force for Alzheimer's, and that drugs meant to prevent dementia in humans could be tested using this model.[12] Several drugs have been developed to target tau, but most have had terrible side effects.

Instead of having a drug that can turn off phosphorylation in tau proteins, the technology developed by Photys could modify the phosphorylation, potentially avoiding the side effects caused by the blockers and inhibitors while preventing the tau proteins from accumulating and tangling. While there are no current estimates on the cost to produce these drugs, it would likely be much higher than the simpler inhibitors and blockers. No matter how much the consumer is charged, they would have no choice: there is no alternative. If the drugs cost twice as much as the most expensive inhibitor on the market, and around 65% of Americans with Alzheimer's have the accumulation tau proteins with 10% of those people taking the drug, that would produce around 408 billion dollars in revenue annually.[14]

Another potential use of Photys is in diabetes. The founder of Photys has stated that a potential use of the technology could be used to cure type-1 diabetes (TD1) by inducing phosphorylation on insulin receptors.[15] Both TD1 and type-2 diabetes (TD2) are associated with hyperphosphorylation of the tau protein, the same protein that is the likely cause of most Alzheimer's disease cases.[16] Meaning the same drugs developed for Alzheimer's patients could be given to patients with TD1 and TD2 to reduce cognitive decline in patients and prevent Alzheimer's disease from presenting symptoms, which is much more likely to occur in diabetic patients. However, it is most viable for glucokinase-maturity-onset-diabetes of the young (GCK-MODY) which is a form of diabetes caused from the mutation of one gene. GCK-MODY impacts less than 5% of all diabetes, and those with GCK-MODY typically don't need any treatment anyways.[17] However, if 0.1% of all diabetic patients in the US took the same drugs, using the same assumptions, a revenue of 405 billion dollars could be made (in addition to the revenue from Alzheimer's patients) yearly.

Most of the diseases with identified genes are forms of cancers, which could also be treated with Photys technology. With cancers, there are other options for treatment, and a different drug would have to be formulated for each type of cancer or potentially each person's cancer. The future of Photys in the cancer sector is much more unclear. Even without using the technology for cancer treatments, a potential 800 billion dollars of revenue per year in the US is much higher than anything else. The entirety of North America produced about 8.81 billion dollars in revenue in 2023 for insulin.[18] Photys has the potential to produce the biggest money makers in the therapeutics industry.

Risks and Considerations

There are a few factors that must be considered when discussing Photys and PHICS technology, both when comparing PHICS with traditional phosphorylation therapies and when considering PHICS as a standalone drug. The main phosphorylation-based therapy right now utilizes tyrosine kinase inhibitors (TKIs) that modulate specific kinase activity to influence a number of factors.[19] The main risk associated with TKI therapeutics is drug toxicity, as TKIs can

adversely affect a large number of the body's organs.[19] Beyond that, TKIs have been found to produce resistance mutations within the body, primarily occurring in BCR-ABL1 related mechanisms.[20] As mentioned above, PHICS differs from traditional phosphorylation therapies in that it can specifically target a desired protein, regardless of whether or not the protein is a natural kinase substrate.[1] Because of this, PHICS introduces a set of benefits and risks compared to traditional therapies that must be taken into consideration when discussing the efficacy of PHICS.

The main risks presented by PHICS compared to traditional therapies are not fully explored, but current publications on the technology reveal some possible risks that are not found in traditional phosphorylation therapies. In a recent publication, initial testing of PHICS revealed a couple problems that could present themselves as risks in the future. These main problems included the appearance of a "hook effect" during ternary complex formation, negative cooperativity between BRD4 and ABL, and a catalytic turnover dependency.[2] However, these issues are not necessarily all bad, as most bifunctional molecules have these same traits.[1] For example, most proteolysis-targeting chimeras (PROTACs) are susceptible to the aforementioned hook effect, a phenomenon in which drug effectiveness begins to decrease at high concentrations.[21] The reason that these traits in PHICS could be issues are the same as for other drugs; the existence of the hook effect and catalytic turnover introduce the need for proper diligence and testing of PHICS to ensure that drug is able to work as effectively as possible. Catalytic turnover specifically presents itself as somewhat of a "double-edged sword," as turnover increases the efficiency of the drug, but could also make the drug difficult to control, possibly leading to unwanted effects like toxicity and the disturbance of regular cellular processes. The negative cooperativity that arises between BRD4 and ABL in the presence of PHICS (**Fig. 2**)[2] is also a low-risk issue, as it really only would decrease phosphorylation efficacy rather than introduce any health risks to the body. More risks are sure to appear during clinical testing, but most of the main possible risks presented by PHICS will be worked out in that time with due diligence.[15]

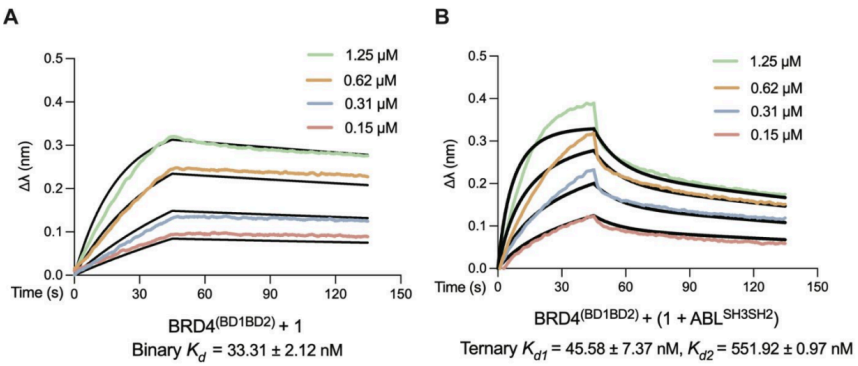


Figure 2. *Binding kinetics for the binary and ternary complexes. (A) Binary BLI sensogram of the BRD4^{BD1BD2} binding to PHICS 1. (B) Ternary complex BLI sensogram of a binary ABLSH3SH2-PHICS 1 complex binding to BRD4^{BD1BD2}.*

Financially, PHICS presents a bit larger of a risk. As mentioned, it is estimated that PHICS could cost upwards of $250,000 per person each year, on average. However, it was recently found that PHICS can be successfully derived from existing kinase inhibitors rather than the previously used non inhibitory allosteric binders.[2] This discovery could cut the cost of PHICS manufacturing for Photys by a large margin, as PHICS could be derived directly from pre-existing scaffolds rather than spending both time and money to synthesize an entirely new molecule.[2] Through this, the scalability of PHICS production and research looks much more positive compared to initial projections.

Overall Recommendation

After a thorough qualitative analysis of the company's standing and potential, we at RFS BioConsulting consider Photys Therapeutics a strong investment. We understand that therapeutics are incredibly volatile and there is significant risk moving forward into clinical trials and FDA approval, but believe that small molecule therapies, such as PHICS, are the future of medicine. Photys is backed by major corporate companies such as Eli Lily and Co. and Merck investing $75 million USD during series-A funding, with underlying support and belief throughout the pharmaceutical industry. Coupling their transformation technology of synthetic PHICs with an experienced and savvy leadership team, we believe that Photys is a very strong investment. Recently appointed Chief Executive Officer, Brian Felton, has previously navigated a deal to sell an mRNA therapeutic company, Translate Bio, to Sanofi for $3.2 Billion USD.[22] The strong scientific backing of proximity-induced phosphorylation opens the door to many of the largest markets within medicine, including cancer, diabetes, and Alzheimer's. At RFS BioConsulting, we believe that Photys will be a disruptor in the therapeutic field and will prove to be a successful investment moving forward.

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